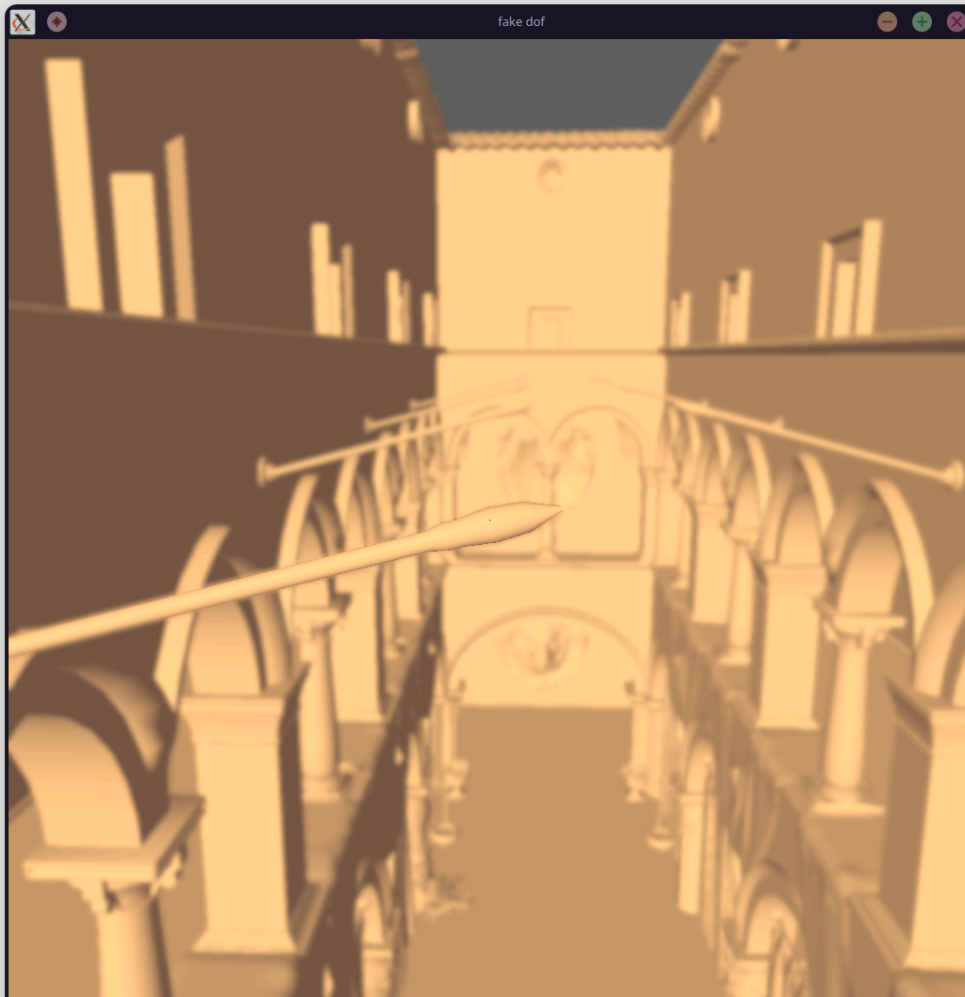
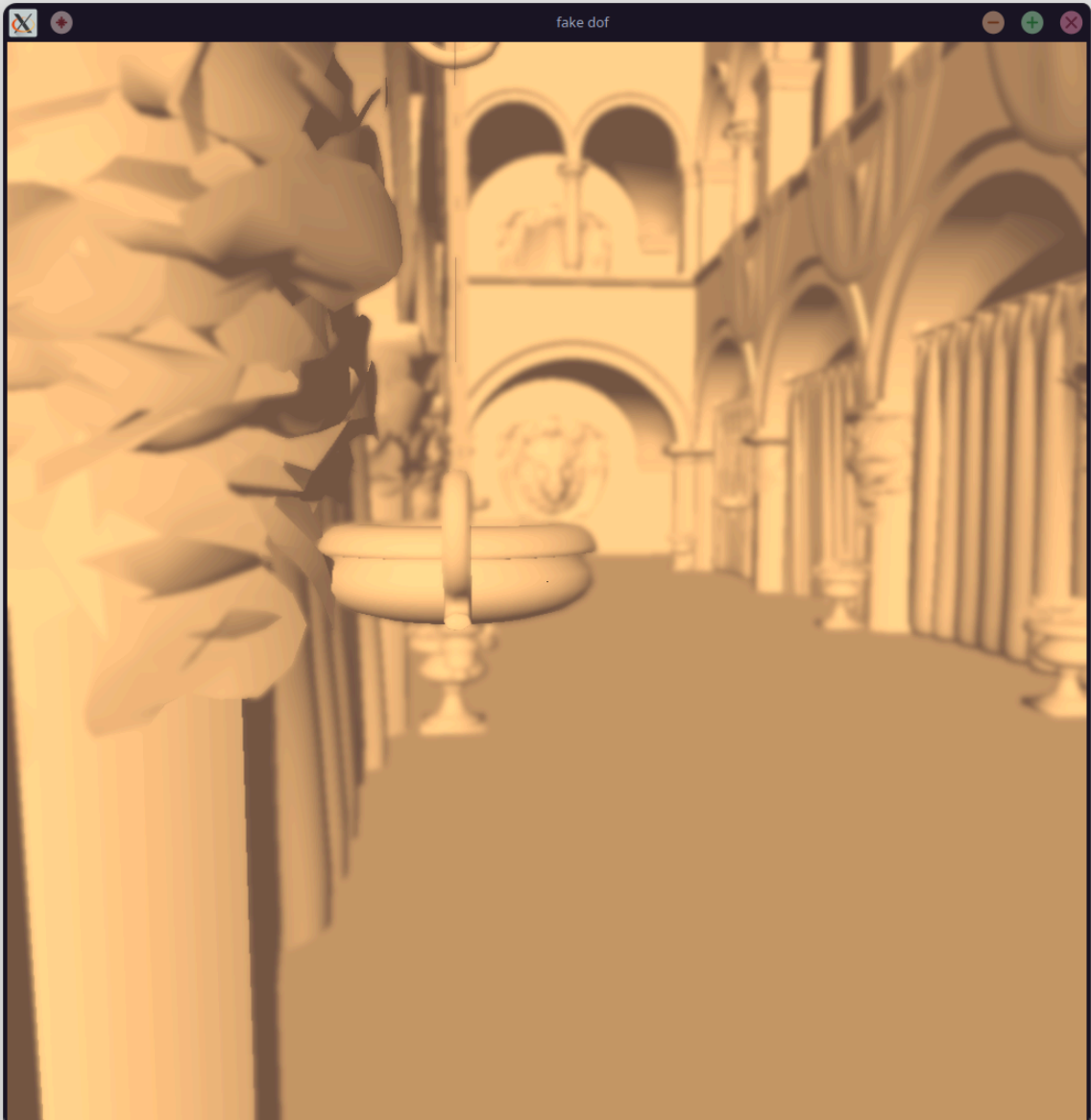


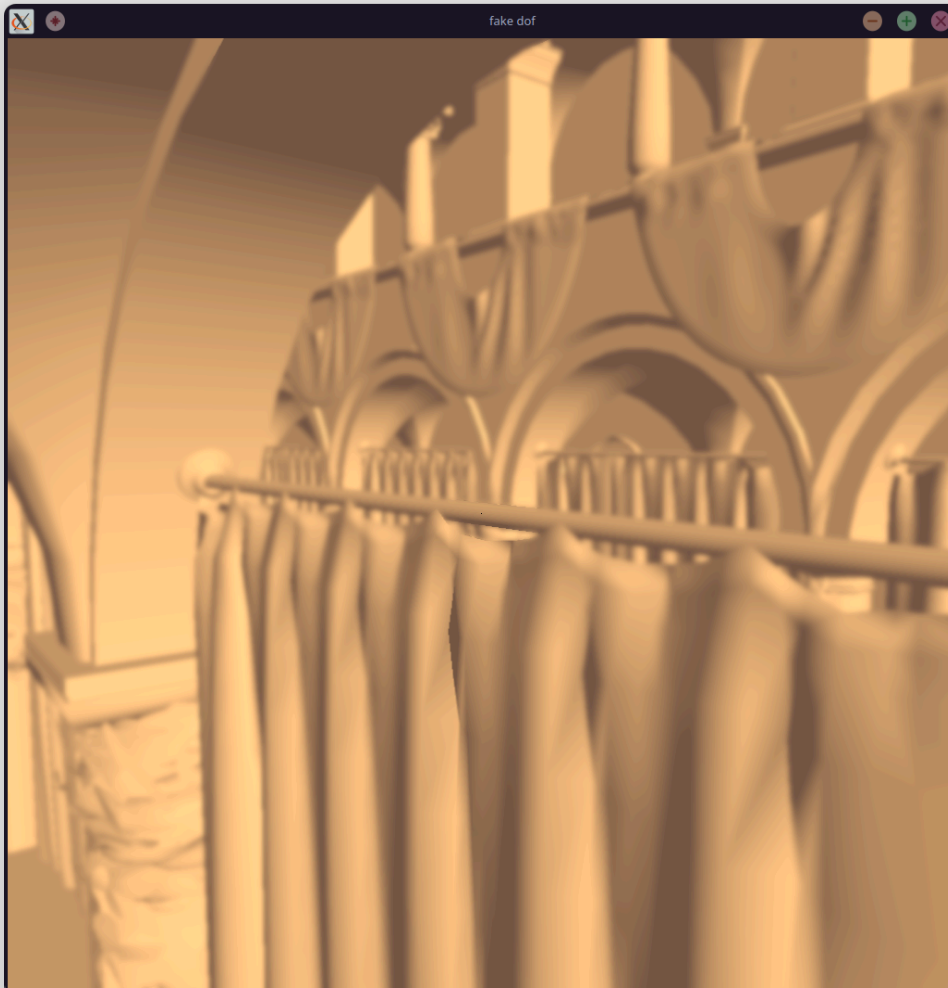
Fake Depth of Field Utsawb Lamichhane

For my final project I took the techniques we learned in class, and created a fake depth of field post processing effect in a Vulkan compute shader

<https://github.com/Utsawb/letc/tree/rewrite>







- Firstly I wrote a Vulkan renderer that goes through all the verbose steps to get models rendered to the screen.
- Then I used tinyglTF to load in the famous Sponza scene
- Then I used a Gaussian Kernel to blur the scene per pixel, and adjust the strength of the blur based on the distance between the center pixel and the rendered pixel. This simulates the focal plane that traditional cameras have, without the expensive calculations required via ray tracing.
- During testing I noticed that due to the non-linear nature of the depth buffer that the values I picked were too small, so I converted the depth values from the buffer back into a linear scale, so that I may work with them.
- Then, after skipping values near the target focal plane, I blur them based on the depth buffer.
- Although this effect caused some haloing near the edge, this is a tried and true method of simulating some depth of field for a very low computational cost.